



# ELSTE Data science

*Some concept of Python coding and statistical inference*

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# Get ready

- Go to <https://elste-master.github.io/Data-Science/>
  - **Download slides:** Class 1 > Overview
  - **Setup VS Code:** Appendices > Setup VS Code

## ELSTE Data Science Course



ELSTE Data Science Master Course

Class 1: Pandas >

Class 2: Descriptive stats >

Appendices v

Setup VS Code

Markdown for Jupyter

Appendices > Setup VS Code

## Setup VS Code

This tutorial guides you through the installation and configuration of Visual Studio Code (VS Code) for Python development in the 0013 computer room at the University of Geneva.

## Open VS Code

1. Create a folder on your **H:/** drive named **data\_science**
2. Open **VS Code**

### On this page

Setup VS Code

Open VS Code

Create a new Jupyter Notebook

Install the Python extensions

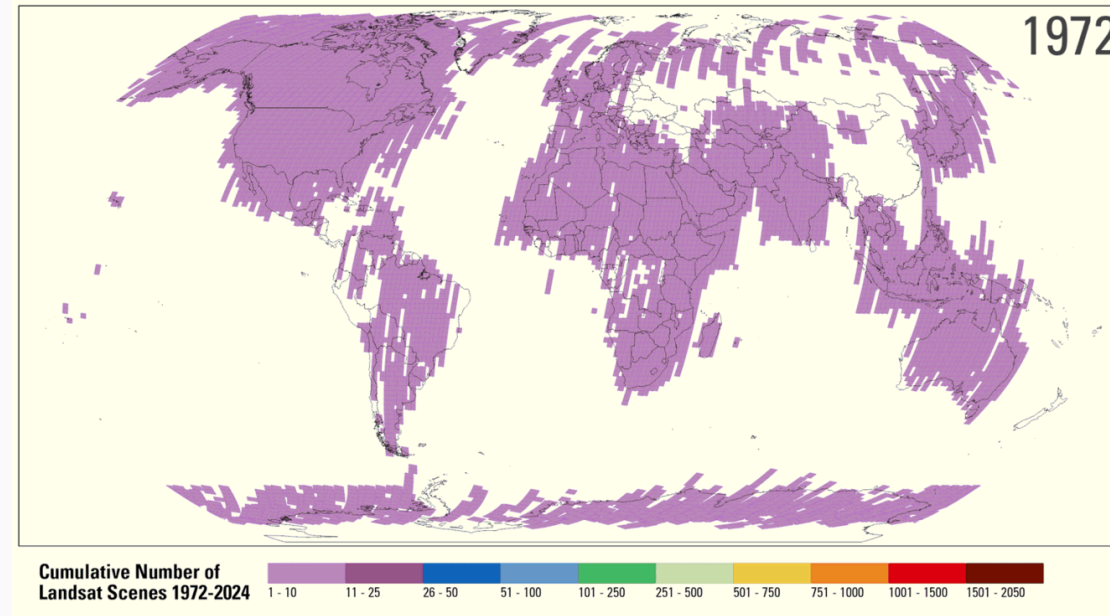
Install Jupyter

Select the right Python

Open the course material in VS Code

# Why this class?

*Data deluge: The big EO data landscape*



- **Landsat:**
  - Download: 53 images/day (2001) - 220'000 images/day (2017)
  - 5 million images of the Earth surface → > 5 PB
- **ESA Sentinel 1 & 2** → 4.6TB /daily!

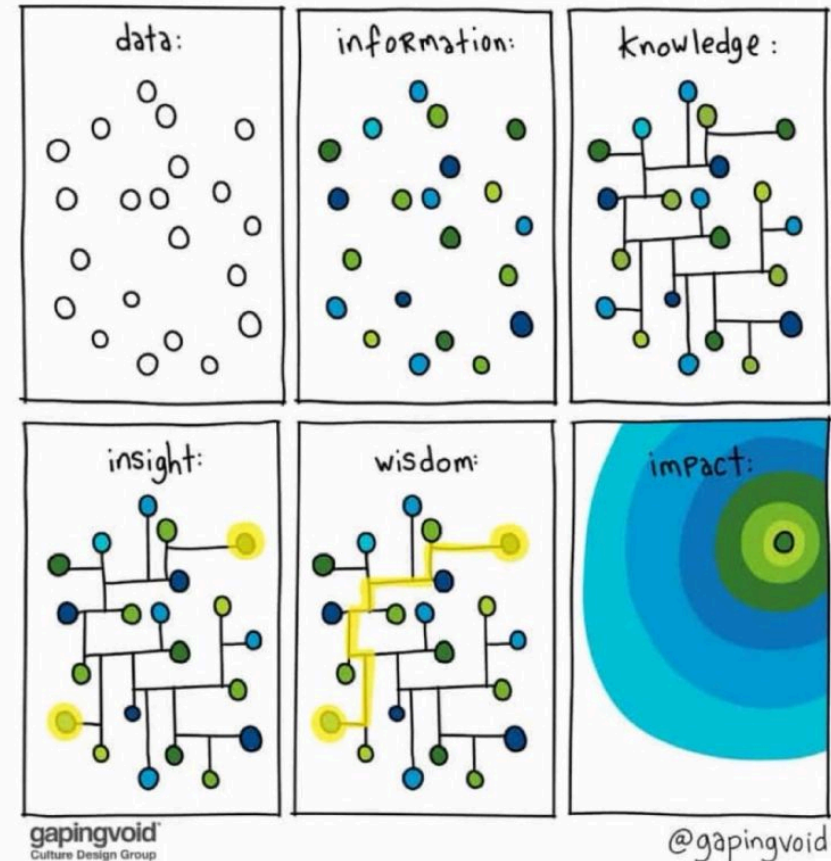
# Why this class?

As scientists, we are exposed to:

- Increasing computational power/facilities
- Increasing amount of data

How do we make sense of it all?

- Scientific coding → gateway to scientific data analysis
- Data science → extracting meaningful information and insights from data



# What is data science

*Wikipedia*

Data science is an interdisciplinary academic field that uses statistics, scientific computing, scientific methods, processing, scientific visualization, algorithms and systems to extract or extrapolate knowledge from [...] data.

*National Institutes of Standards and Technology*

The field that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data.

# Class schedule

- **Fall part at UNIGE:** 6 × 3-h long sessions
- **Assistants:** Simon Thivet, Filippo Salmaso, Lionel Voirol
- **Format:** Theory, live coding and exercises

*October 15 and 22*

- **Seb** → Intro to data science libraries for Python → Pandas, seaborn

*October 29 and November 5*

- **Stéphane** → Intro to statistical method for data inference

*November 12 and 19*

- Flexible as a function of how we progress
- Exam format to be defined



# Class objectives

- Scientific coding and stats often appear daunting
- True, there can be a steep learning curve

## *Objectives*

- **Not** to make you an expert developer / statistician **but**
  - To help you gain the confidence that you are totally capable of doing it
  - To make you realise that what you will get in terms of research capabilities is worth the effort!



# Get involved!!!

We want this class to be useful for **your research!**

Try to contextualise the course material to **your research**:

- Do you already have your own **datasets** you could bring to class?
- Can you use the course to **formulate new research questions?**
- Do you know any open-access datasets relevant to your fields?
  - If not, can you find some for the next weeks?





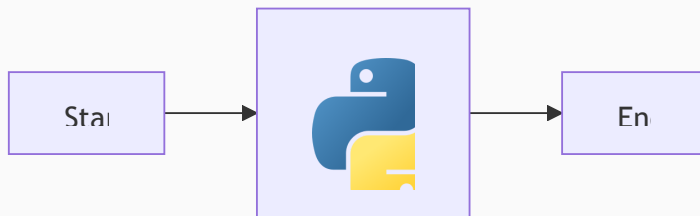
# Why coding?

# Motivations

- **Example:** A common - but unnecessarily complicated - workflow of many specialised softwares



- **Objective:** Integrate full research workflow in an environment supporting all tasks



# Motivations

## *Motivation 1: Automation*

### Cotopaxi volcano

- Reconstructing eruption source parameters (ESP) from tephra deposits
  - *How do different measurement methods influence ESP estimates?*
- 40 outcrops
- ~10 samples per outcrop
- ~7 measurement methods per sample

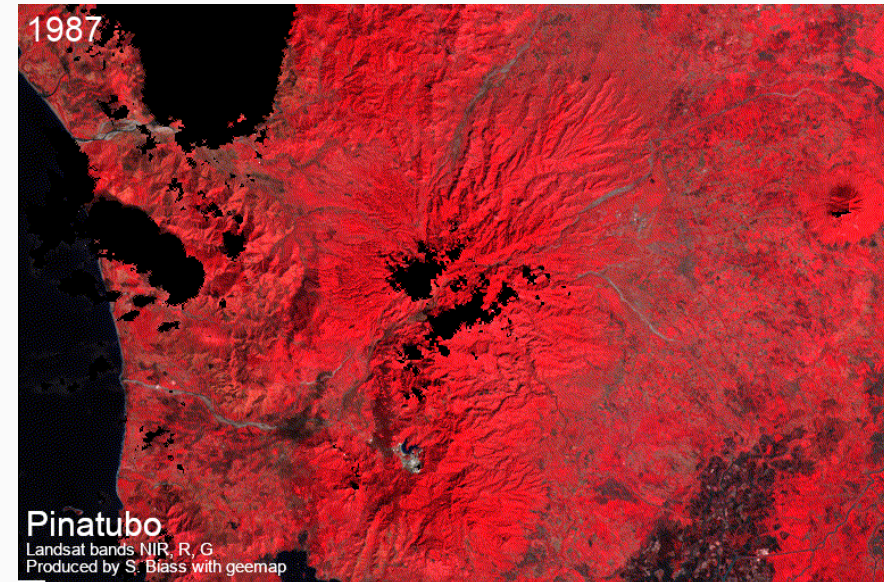


# Motivations

## *Motivation 2: Data analysis*

**Example 1:** Exploit catalogues of big Earth Observation data

- **Revisit big EO catalogues to infer new knowledge**
  - *What controls the impact and recovery of vegetation following eruptions?*

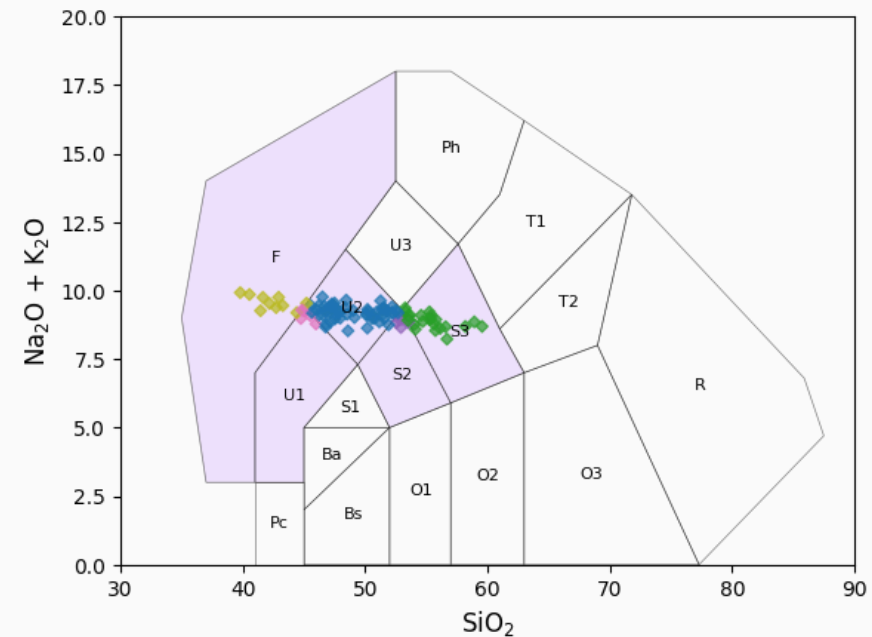


# Motivations

## *Motivation 2: Data analysis*

**Example 2:** Streamline global geochemical analyses

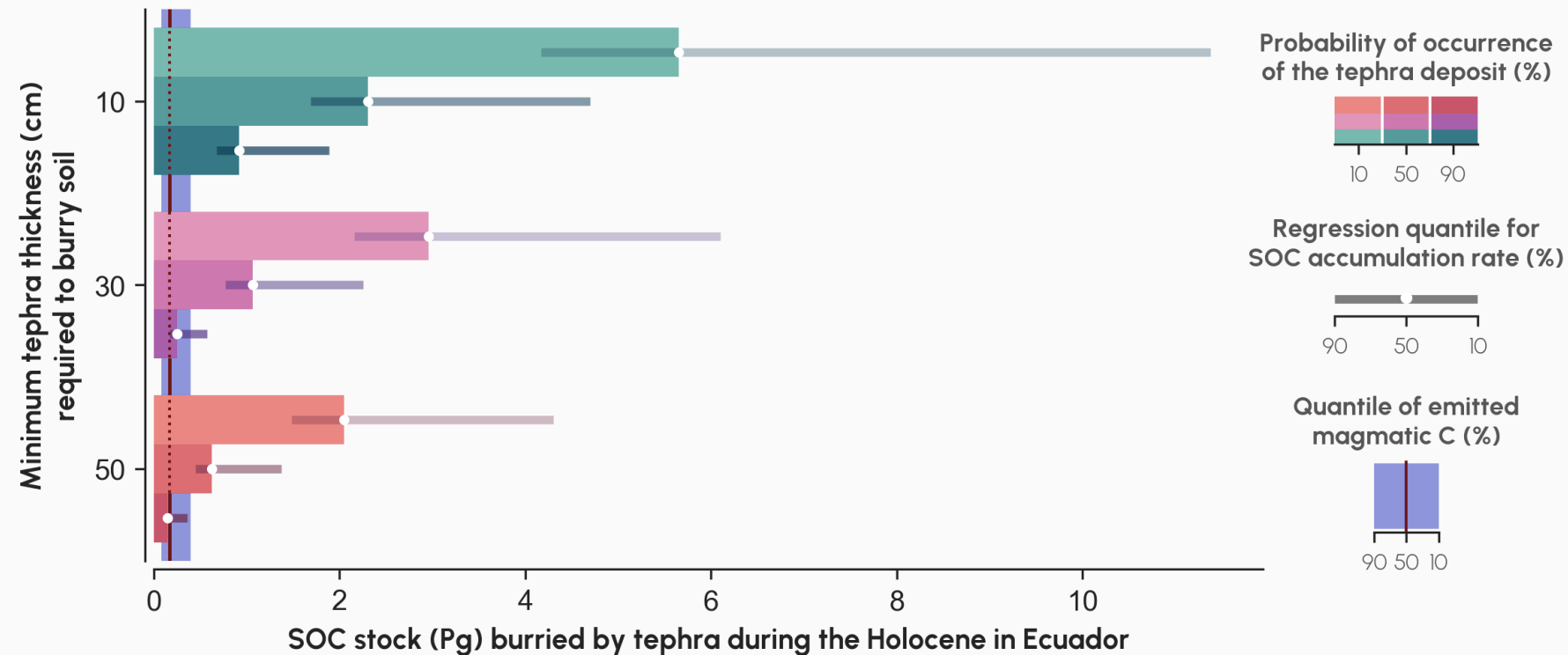
- **Access global databases**
  - e.g., *Georock* database
- **Automatic dedicated analyses/plots**
  - e.g., *TAS* diagrams and classification using `pyrolite`



# Motivations

## *Motivation 3: Visualisation*

**Example 1:** In volcanically-active regions, soil burial after explosive eruptions capture more carbon that they emit.

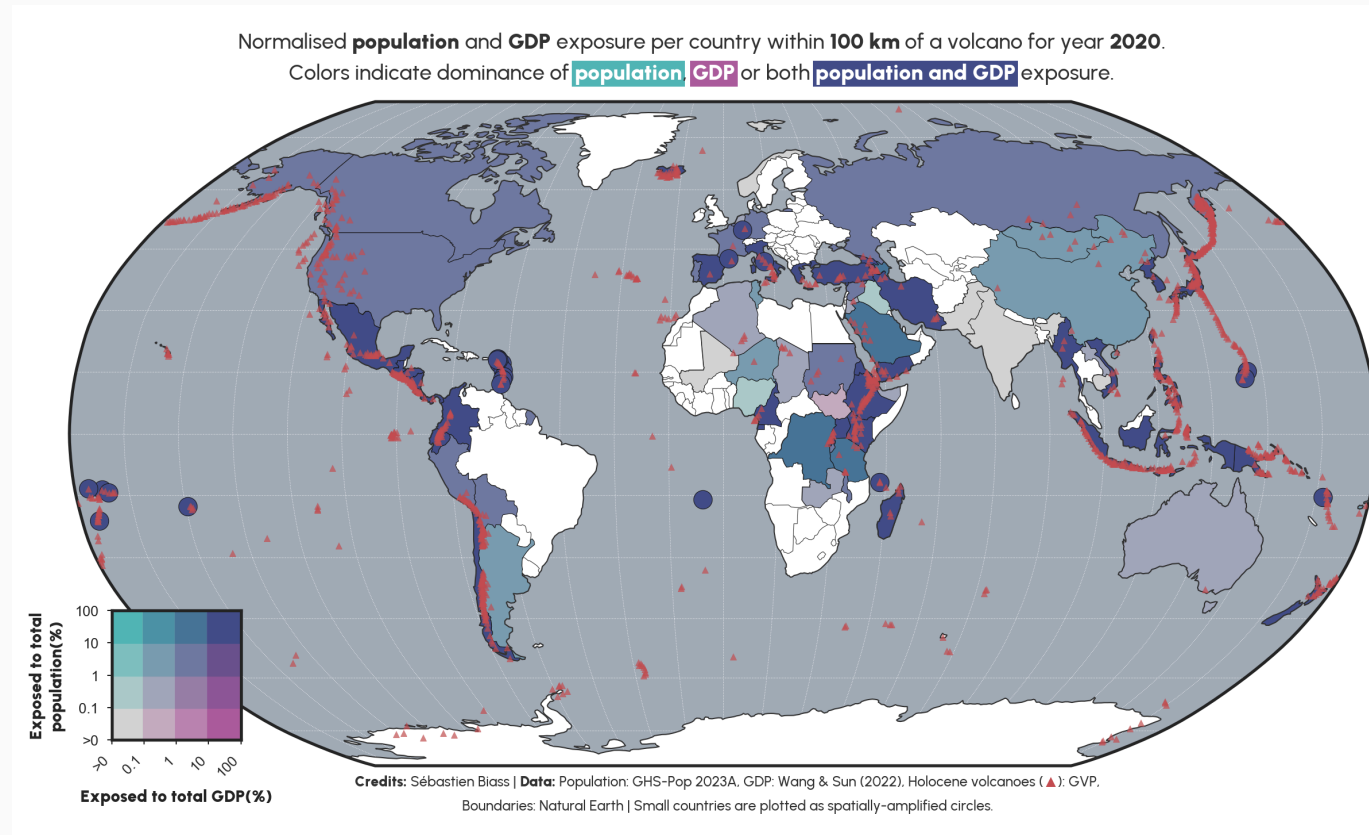




# Motivations

## *Motivation 3: Visualisation*

**Example 2:** In Small Island States,  $\geq 90\%$  of population and GDP is exposed to volcanic hazards.





# Why Python?





# Python's advantages <sup>1</sup>

## 1. Free

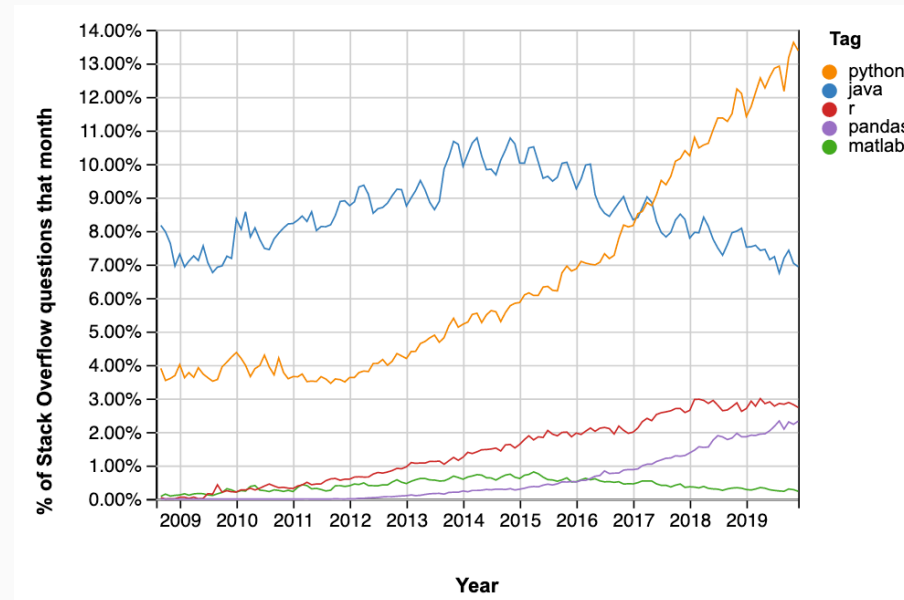
## 2. Open source

- Not dependent on any company
- Large online community

## 3. Old → stable (started in 1991)

## 4. **Very** popular in science

## 5. Relatively easy to learn



1. Inspired by the [School of Oceanography at University of Washington](#)



# What is Python

## High-level computing language

- Compiled → doesn't directly speak to the computer, but is *interpreted* by another language
  - Usually slower, but much, much easier to use!

## Modular

- Python is composed of a core of functions complemented by wide ecosystem of specialized packages for specific tasks
  - A function is a bit of code to perform a **specific task**
  - A module is a collection of **functions**
  - A package is composed by multiple **modules**



# An example of packages, modules and functions

In the code below, we first load the necessary packages and libraries:

- `matplotlib` is the main **visualisation** package used in Python
- `pyplot` is a module of `matplotlib` that provides easy-to-use functions for **plotting** data
- `figure` is a function of `pyplot` and is the main function to prepare a plot

```
1 # Import the packages and modules
2 from matplotlib import pyplot as plt
3 # Set up a figure for plotting
4 plt.figure(...)
```



# How to use Python

## Option 1: Your own computer

- Requires some setup
- No internet connection required
- Slow if computer is slow
- All data stored on your own computer

### ! Environment manager!

Make sure to install an **environment manager** → e.g. **Miniconda**

## Option 2: In the cloud → Google Collab

- No setup required
- No internet connection required
- Decent speed, free as long as Google says it is free
- All data stored in the cloud



# How to run Python

## 3 main ways to run Python...

1. Run Python `.py` scripts from the command line → *deprecated*
2. Run blocks of code from within a Python `.py` script using `ipykernel` → **similar behaviour to R/Matlab** = ❤️
3. Use **Jupyter Notebooks** in `.ipynb` files

## ...in two different environments

1. Dedicated software: **VSCode**, **PyCharm**, **Spyder**
2. Web browser: **JupyterLab**



# Start coding!

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